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ASHRAE is the world's foremost technical society in the fields of heating, ventilation, air conditioning, and refrigeration. Its members worldwide are individuals who share ideas, identify needs, support research, and write the industry's standards for testing and practice. The result is that engineers are better able to keep indoor environments safe and productive while protecting and preserving the outdoors for generations to come.

One of the ways that ASHRAE supports its members' and industry's need for information is through ASHRAE Research. Thousands of individuals and companies support ASHRAE Research annually, enabling ASHRAE to report new data about material

properties and building physics and to promote the application of innovative technologies.

Chapters in the ASHRAE Handbook are updated through the experience of members of ASHRAE Technical Committees and through results of ASHRAE Research reported at ASHRAE conferences and published in ASHRAE special publications, *ASHRAE Transactions*, and ASHRAE's journal of archival research, *Science and Technology for the Built Environment*.

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Preface

The 2017 *ASHRAE Handbook—Fundamentals* covers basic principles and data used in the HVAC&R industry. The ASHRAE Technical Committees that prepare these chapters provide new information, clarify existing content, delete obsolete materials, and reorganize chapters to make the Handbook more understandable and easier to use. An accompanying CD-ROM contains all the volume's chapters in both I-P and SI units.

This edition includes a new chapter:

- **Chapter 36**, Moisture Management in Buildings, presents data on indoor vapor release and measured indoor/outdoor vapor pressure/concentration differences, and discusses moisture sources and sinks that can reduce materials' durability, as well as the negative effects of insufficient or excessive indoor relative humidity.

Other selected highlights include the following:

- **Chapter 7**, Fundamentals of Control, has new content on thermostatic valve actuators, placement of sensors, auxiliary control devices, and network architecture.
- **Chapter 9**, Thermal Comfort, has new content from ASHRAE research project RP-1504 on nonwestern clothing; combined chilled-ceiling, displacement ventilation, and vertical radiant temperature asymmetry effects on sedentary office work; and updates to align with ASHRAE Standard 55-2013.
- **Chapter 10**, Indoor Environmental Health, has updates on bio-aerosols, plus new content on electronic cigarettes and on climate change.
- **Chapter 11**, Air Contaminants, has new content on particle sizes and settling times, particulate contaminant effects, polymerase chain reaction (PCR) measurement, volatility, mercury, e-cigarettes, and 3D printers.
- **Chapter 14**, Climatic Design Information, includes new data for 8118 locations worldwide—an increase of 1675 locations from the 2013 edition of the chapter—as a result of ASHRAE research project RP-1699.
- **Chapter 15**, Fenestration, has updated discussion on U-factor, solar-optical glazing properties, complex glazings and window coverings, tubular daylighting devices (TDDs), and spectrally selective glazing.
- **Chapter 16**, Ventilation and Infiltration, has been updated and revised for clarity throughout, including recent research results on envelope air leakage.
- **Chapter 17**, Residential Cooling and Heating Load Calculations, has updates for 2017 climate data and current standards.
- **Chapter 18**, Nonresidential Cooling and Heating Load Calculations, has new design data for lighting power densities, motors, kitchen equipment, LED lighting, walls and roofs, and an updated example calculation.

- **Chapter 19**, Energy Estimating and Modeling Methods, extensively revised, has new sections on method development history, using models, uncertainty, thermal loads and model inputs, envelope components, HVAC components, terminal components, low-energy systems, natural and hybrid ventilation, daylighting, passive heating, hybrid inverse method, and model calibration.
- **Chapter 20**, Space Air Diffusion, has new content on outlet types and characteristics, air curtains, thermal plumes, and air movement in occupied zones.
- **Chapter 21**, Duct Design, was reorganized for ease of use, and updated for data from the latest version of the *ASHRAE Duct Fitting Database*.
- **Chapter 22**, Pipe Design, has a new title and now incorporates the content of its sister chapter, Pipes, Tubes, and Fittings, from *HVAC Systems and Equipment*. Also added are content on PEX pipe, plus expanded applications.
- **Chapter 24**, Airflow Around Buildings, has new content on flow patterns around building groups and isolated buildings, environmental impacts, pollutant dispersion and exhaust reentrainment, pedestrian wind comfort and safety, and wind-driven rain.
- **Chapter 30**, Thermophysical Properties of Refrigerants, has new or revised data for R-1233zd(E), R-245fa, R-1234ze(E), and R-1234yf.
- **Chapter 34**, Energy Resources, has extensive updates for new statistics on worldwide energy use and resources.
- **Chapter 35**, Sustainability, has new content on the water/energy nexus, embodied energy, and climate change.

This volume is published, as a bound print volume and in electronic format on CD-ROM and online, in two editions: one using inch-pound (I-P) units of measurement, the other using the International System of Units (SI).

Corrections to the 2014, 2015, and 2016 Handbook volumes can be found on the ASHRAE website at www.ashrae.org and in the Additions and Corrections section of this volume. Corrections for this volume will be listed in subsequent volumes and on the ASHRAE website.

Reader comments are enthusiastically invited. To suggest improvements for a chapter, **please comment using the form on the ASHRAE website** or, using the cutout page(s) at the end of this volume's index, write to Handbook Editor, ASHRAE, 1791 Tullie Circle, Atlanta, GA 30329, or fax 678-539-2187, or e-mail mowen@ashrae.org.

Mark S. Owen
Editor

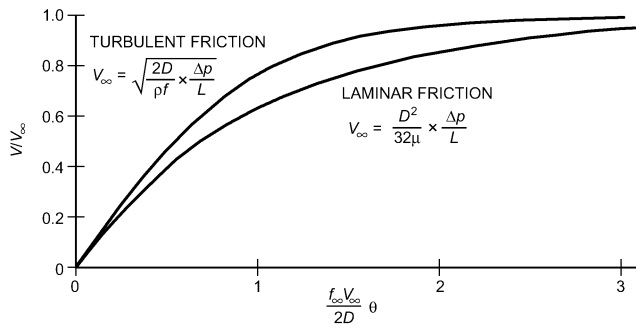


Fig. 20 Temporal Increase in Velocity Following Sudden Application of Pressure

steady-state result for accelerating flows, but appreciably less for decelerating flows. If the friction factor is approximated as constant,

$$\frac{dV}{d\theta} = \frac{\Delta p}{\rho L} - \frac{fV^2}{2D} = A - BV^2 \quad (50)$$

and for the accelerating flow,

$$\theta = \frac{1}{\sqrt{AB}} \tanh^{-1} \left(V \sqrt{\frac{B}{A}} \right) \quad (51)$$

or

$$V = \sqrt{A/B} \tanh(\theta \sqrt{AB}) \quad (52)$$

Because the hyperbolic tangent is zero when the independent variable is zero and unity when the variable is infinity, the initial ($V=0$ at $\theta=0$) and final conditions are verified. Thus, for long times ($\theta \rightarrow \infty$),

$$V_{\infty} = \sqrt{A/B} = \sqrt{\frac{\Delta p / \rho L}{f_{\infty} / 2D}} = \sqrt{\frac{\Delta p (2D)}{\rho L (f_{\infty})}} \quad (53)$$

which is in accord with Equation (30) when f is constant (the flow regime is the fully rough one of Figure 13). The temporal velocity variation is then

$$V = V_{\infty} \tanh(f_{\infty} V_{\infty} \theta / 2D) \quad (54)$$

In Figure 20, the turbulent velocity start-up result is compared with the laminar one, where initially the turbulent is steeper but of the same general form, increasing rapidly at the start but reaching V_{∞} asymptotically.

Compressibility

All fluids are compressible to some degree; their density depends somewhat on the pressure. Steady liquid flow may ordinarily be treated as incompressible, and incompressible flow analysis is satisfactory for gases and vapors at velocities below about 20 to 40 m/s, except in long conduits.

For liquids in pipelines, a severe pressure surge or water hammer may be produced if flow is suddenly stopped. This pressure surge travels along the pipe at the speed of sound in the liquid, alternately compressing and decompressing the liquid. For steady gas flows in long conduits, the pressure drop along the conduit can reduce gas density enough to increase the velocity. If the conduit is long enough, the velocity may reach the speed of sound, and the Mach number (ratio of flow velocity to the speed of sound) must be considered.

Some compressible flows occur without heat gain or loss (adiabatically). If there is no friction (conversion of flow mechanical

energy into internal energy), the process is reversible (isentropic) and follows the relationship

$$\begin{aligned} p/\rho^k &= \text{constant} \\ k &= c_p/c_v \end{aligned}$$

where k , the ratio of specific heats at constant pressure and volume, is 1.4 for air and diatomic gases.

When the elevation term gz is neglected, as it is in most compressible flow analyses, the Bernoulli equation of steady flow, Equation (21), becomes

$$\int \frac{dp}{\rho} + \frac{V^2}{2} = \text{constant} \quad (55)$$

For a frictionless adiabatic process,

$$\int_1^2 \frac{dp}{\rho} = \frac{k}{k-1} \left(\frac{p_2}{\rho_2} - \frac{p_1}{\rho_1} \right) \quad (56)$$

Integrating between upstream station 1 and downstream station 2 gives

$$\frac{p_1}{\rho_1} \left(\frac{k}{k-1} \right) \left[\left(\frac{p_2}{p_1} \right)^{(k-1)/k} - 1 \right] + \frac{V_2^2 - V_1^2}{2} = 0 \quad (57)$$

Equation (57) replaces the Bernoulli equation for compressible flows. If station 2 is the stagnation point at the front of a body, $V_2 = 0$, and solving Equation (57) for p_2 gives

$$p_s = p_2 = p_1 \left[1 + \left(\frac{k-1}{2} \right) \frac{\rho_1 V_1^2}{k p_1} \right]^{k/(k-1)} \quad (58)$$

where p_s is the stagnation pressure.

Because the speed of sound of the gas is $a = kp/\rho$ and Mach number $M = V/a$, the stagnation pressure in Equation (58) becomes

$$p_s = p_1 \left[1 + \left(\frac{k-1}{2} \right) M_1^2 \right]^{k/(k-1)} \quad (59)$$

For Mach numbers less than one,

$$p_s = p_1 \frac{\rho_1 V_1^2}{2} \left[1 + \frac{M_1}{4} + \left(\frac{2-k}{24} \right) M_1^4 + \dots \right] \quad (60)$$

When $M = 0$, Equation (60) reduces to the incompressible flow result obtained from Equation (9). When the upstream Mach number exceeds 0.2, the difference is significant. Thus, a pitot tube in air is influenced by compressibility at velocities over about 66 m/s.

Flow Measurement. For isentropic flow through a converging conduit such as a flow nozzle, venturi, or orifice meter, where velocity at the upstream station 1 is small, Equation (57) gives

$$V_2 = \sqrt{\frac{2k}{k-1} \left(\frac{p_1}{\rho_1} \right) \left[1 - \left(\frac{p_2}{p_1} \right)^{(k-1)/k} \right]} \quad (61)$$

The mass flow rate is

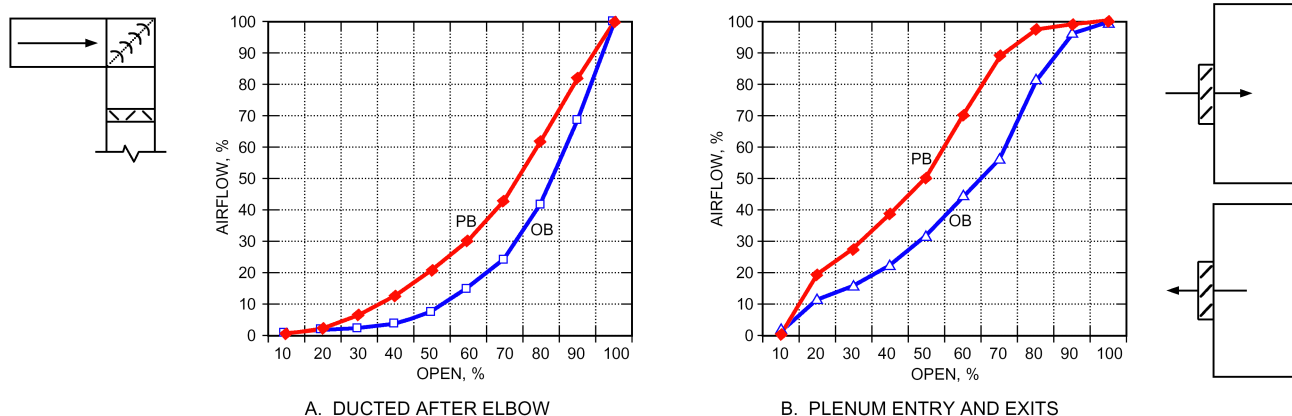


Fig. 15 Inherent Curves for Ducted and Plenum-Mounted Dampers (RP-1157)

Based on data in van Becelaere et al. (2004)

ducted application. These are “inherent” curves, where pressure drop across a damper is held constant as the damper rotates (so it has 100% authority). In real applications, the authority is lower (higher losses of other system components besides the damper). As system pressure losses increase, the curves move up. Note that PB dampers are significantly above linear in most cases.

Figure 15 shows three more applications with PB and OB dampers. The ducted damper has some disturbance and pressure loss ahead of it, to simulate a more realistic situation than those of AMCA 5.3. Nevertheless, the response curves are similar to AMCA 5.3. The plenum entry dampers show irregular results. Again, these are inherent curves, and lower authority causes the curves to move up toward more flow at smaller angles.

The curves shown here are typical, but do not represent every scenario. Thousands of installation variations exist, and slight variations in response always occur. For additional application examples and greater detail, see ASHRAE research project RP-1157 (van Becelaere et al. 2004).

Application. Dampers require engineering to achieve defined goals. A common application is a flow control damper, which modulates airflow. The curves in Figure 13 can be used to pick a damper with a pressure drop and authority that provides near-linear response. Another common application is economizer outdoor air, return air, and exhaust air dampers. Selection of these dampers depends on the system design, as discussed in ASHRAE Guideline 16-2014.

Damper leakage is a concern, particularly where tight shutoff is necessary to significantly reduce energy consumption. Also, outdoor air dampers in cold climates must close tightly to prevent coils and pipes from freezing. Low-leakage dampers cost more and require larger actuators because of friction of the seals in the closed position; however, the energy savings offset the extra cost.

Actuators. Either electricity or compressed air is used to actuate dampers.

Pneumatic damper actuators are similar to pneumatic valve actuators, except that they have a longer stroke or the stroke is increased by a multiplying lever. Increasing air pressure produces a linear shaft motion, which, through a linkage, moves the crank arm to open or close the dampers. Releasing air pressure allows a spring to return the actuator. Double-acting actuators without springs are available also.

Electric damper actuators can be proportional or two-position. They can be spring return or nonspring. The simplest form of control is a floating three-point controller, in which contact closures drive the motor clockwise or counterclockwise. In addition, a variety of standard electronic signals from electronic controllers or

DDC systems, such as 4 to 20 mA, 2 to 10 V (dc), or 0 to 10 V (dc), can be used to control proportionally actuated dampers.

Most modern actuators are electronic and use more sophisticated methods of control and operation. They are inherently positive positioning and may have communications capabilities similar to industrial-process-quality actuators.

A two-position spring-return actuator moves in one direction when power is applied to its internal windings. When no power is present, the actuator returns (via spring force) to its fail-safe (normal) position. Depending on how the actuator is connected, this action opens or closes the dampers. A proportional actuator may also have spring-return action.

Mounting. Damper actuators are mounted in different ways, depending on the size and accessibility of the damper, and the power required to move the damper. The most common method of mounting electric actuators is directly over the damper shaft (direct coupling) with no external linkage. Actuators can also be mounted in the airflow on the damper frame and be linked directly to a damper blade (though this may deform blades and make access for repairs difficult), or mounted outside the duct and connected to a crank arm attached to a shaft extension of one of the blades. Mounting methods that link the actuator to a single blade are not recommended because of the potential to bend the single drive blade or to have the blade-to-blade linkages get out of adjustment.

Large dampers may require two or more actuators, which are usually mounted at separate points on the damper. An alternative is to install the damper in two or more sections, each section being controlled by a single damper actuator. Positive positioners may be required for proper sequencing, as with a small damper controlled to maintain healthy indoor air quality, with a large damper being independently controlled for economy-cycle free cooling.

Pneumatic Positive (Pilot) Positioners

Where accurate positioning of a modulating pneumatic damper or valve is required, use positive positioners. A positive positioner provides up to full supply control air pressure to the actuator for any change in position required by the controller. A pneumatic actuator without a positioner may not respond quickly or accurately enough to small changes in control pressure caused by friction in the actuator or load, or to changing load conditions such as wind acting on a damper blade. A positive positioner provides finite and repeatable positioning change and allows adjustment of the control range (spring range of the actuator) to provide a proper sequencing control of two or more control devices.

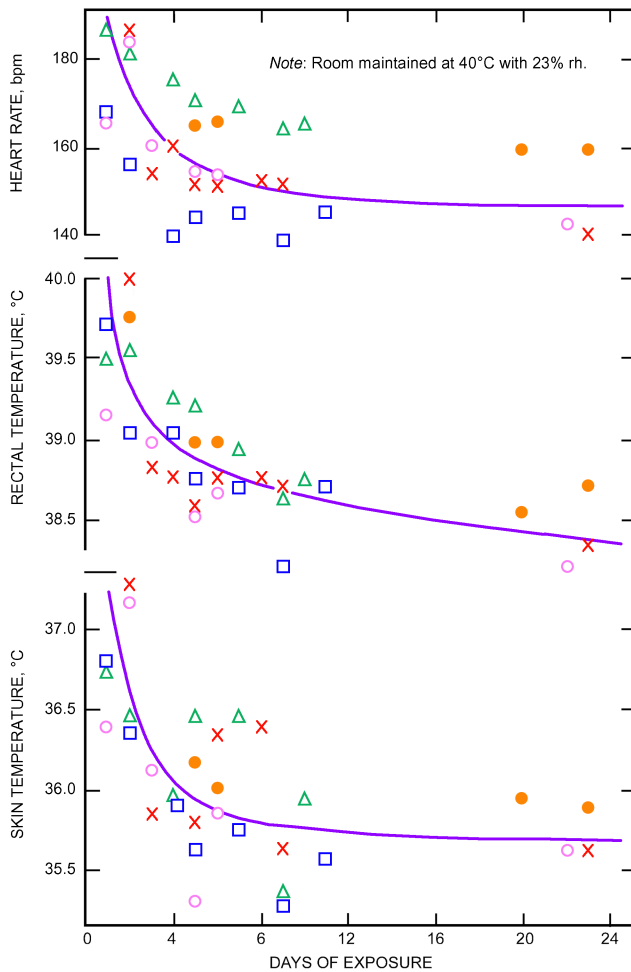


Fig. 26 Acclimatization to Heat Resulting from Daily Exposure of Five Subjects to Extremely Hot Room
(Robinson et al. 1943)

uncomfortable; this represents a drop of approximately 2.6 K (or about 7% of total heat content) in mean body temperature.

This loss can occur during 1 to 2 h of sedentary activity outdoors. A sleeping individual will wake after losing about 314 kJ, decreasing mean skin temperature by about 3 K and deep body temperature by about 0.5 K. A drop in deep body temperature (e.g., rectal temperature) below 35°C threatens a loss of body temperature regulation, and 28°C is considered critical for survival, despite recorded survival from a deep body temperature of 18°C.

Activity level also affects human performance. Subjective sensations reported by sedentary subjects at a mean skin temperature of 33.3°C are comfortable; at 31°C, uncomfortably cold; at 30°C, shivering cold; and at 29°C, extremely cold. The critical subjective tolerance limit (without numbing) for mean skin temperature appears to be about 25°C. However, during moderate to heavy activity, subjects reported the same skin temperatures as comfortable. Although mean skin temperature is significant, the temperature of the extremities is more frequently the critical factor for comfort in the cold. Consistent with this, one of the first responses to cold exposure is vasoconstriction, which reduces circulatory heat input to the hands and feet. A hand-skin temperature of 20°C causes a report of uncomfortably cold; 15°C, extremely cold; and 5°C, painful. Identical verbal responses for the foot surface occur at approximately 1.5 to 2 K warmer temperatures.

An ambient temperature of -35°C is the lower limit for useful outdoor activity, even with adequate insulative clothing. At -50°C, almost all outdoor effort becomes exceedingly difficult; even with appropriate protective equipment, only limited exposure is possible. Reported exposures of 30 min at -75°C have occurred in the Antarctic without injury.

In response to extreme heat loss, maximal heat production becomes very important. When the less-efficient vasoconstriction cannot prevent body heat loss, shivering is an automatic, more efficient defense against cold. This can be triggered by low deep body temperature, low skin temperature, rapid change of skin temperature, or some combination of all three. Shivering is usually preceded by an imperceptible increase in muscle tension and by noticeable gooseflesh produced by muscle contraction in the skin. It begins slowly in small muscle groups, initially increasing total heat production by 1.5 to 2 times resting levels. As body cooling increases, the reaction spreads to additional body segments. Ultimately violent, whole-body shivering causes maximum heat production of about 6 times resting levels, rendering the individual totally ineffective.

Given sufficient cold exposure, the body undergoes changes that indicate cold acclimatization. These physiological changes include (1) endocrine changes (e.g., sensitivity to norepinephrine), causing nonshivering heat production by metabolism of free fatty acids released from adipose tissue; (2) improved circulatory heat flow to skin, causing an overall sensation of greater comfort; and (3) improved circulatory heat flow to the extremities, reducing the risk of injury and allowing activities at what ordinarily would be severely uncomfortable temperatures in the extremities. Generally, these physiological changes are minor and are induced only by repeated extreme exposures. Nonphysiological factors, including training, experience, and selection of adequate protective clothing, are more useful and may be safer than dependence on physiological changes.

Food energy intake requirements for adequately clothed subjects in extreme cold are only slightly greater than those for subjects living and working in temperate climates. This greater requirement results from added work caused by (1) carrying the weight of heavy clothing (energy cost for heavy protective footwear may be six times that of an equivalent weight on the torso); and (2) the inefficiency of walking in snow, snowshoeing, or skiing, which can increase energy cost up to 300%.

To achieve proper protection in low temperatures, a person must either maintain high metabolic heat production by activity or reduce heat loss by controlling the body's microclimate with clothing. Other protective measures include spot radiant heating, showers of hot air for work at a fixed site, and warm-air-ventilated or electrically heated clothing. Extremities (e.g., fingers and toes) are at greater risk than the torso because, as thin cylinders, they are particularly susceptible to heat loss and difficult to insulate without increasing the surface for heat loss. Vasoconstriction can reduce circulatory heat input to extremities by over 90%.

Although there is no ideal insulating material for protective clothing, radiation-reflective materials are promising. Insulation is primarily a function of clothing thickness; the thickness of trapped air, rather than fibers used, determines insulation effectiveness.

Protection for the respiratory tract seems unnecessary in healthy individuals, even at -45°C. However, asthmatics or individuals with mild cardiovascular problems may benefit from a face mask that warms inspired air. Masks are unnecessary for protecting the face because heat to facial skin is not reduced by local vasoconstriction, as it is for hands. If wind chill is great, there is always a risk of cold injury caused by freezing of exposed skin. Using properly designed torso clothing, such as a parka with a fur-lined hood to minimize wind penetration to the face, and 10 W of auxiliary heat to each hand and foot, inactive people can tolerate -55°C with a 16 km/h wind for more than 6 h. As long as the skin temperature of fingers remains

be appropriate in any particular application. Collection directly on **filter paper** is simple and direct, but may dehydrate some organisms and underestimate exposure for live counting techniques. **Glass impingers** are an effective and standard method, but may overestimate exposure because liquid contact and agitation can break clusters into smaller individual organisms, which are then each counted as a separate entity. **Slit-to-agar samplers** may give a more accurate culturable colony count, but do not measure non-culturable organisms or fragments, parts, or components. In general, culture plate impactors, including multiple- and single-stage devices as well as slit-to-agar samplers, are most useful in office environments where low concentrations of bacteria and fungi are expected. Some multihole impactors require application of a positive hole correction factor to the raw counts to compensate for multiple organisms focused aerodynamically and landing in the same place on the media. Because not all microorganisms can grow on the same media, impactors that separate samples must be collected for each. Liquid impingement subculturing allows plating one sample on multiple media. Filter cassette samplers are useful for some hardy microorganisms or components (e.g., endotoxins) and allergen analyses. Filter cassettes can also be used for spore counts.

Nonculture methods for fungal spores and pollen grains generally involve exposing an adhesive-coated glass slide or plate for a specific time period, then counting calibrated areas under the microscope, and calculating the number in a measured volume of air. Measurement methods for pollen are not discussed further here, because data are widely available in the public domain.

Some viruses, bacteria, algae, and protozoa are more difficult to culture than fungi, and air-sampling methodology for these organisms may not be practical. For example, *Legionella* requires special nutrients and conditions for growth, and thus may be difficult to recover from air. To further complicate the issue, not all fungi grow on any one media, so media selection may be important.

Rank-order assessment is used to interpret air-sampling data for microorganisms (Macher 1999). Individual organisms are listed in descending order of abundance for a complainant indoor site and for one or more control locations. The predominance of one or more microbes in the complainant site, but not in the control sites or outdoors, suggests the presence of a source for that organism. An example is shown in Table 4.

Recently, quantitative real-time **polymerase chain reaction (PCR)** (Mullis 1990) testing has been developed for routine diagnosis of specific microorganisms in clinical microbiology laboratories (Espy 2006) and for mold determination in residential and commercial venues. PCR is a molecular technique for multiplying a specific strand of DNA or RNA millions of times by manipulating the chemical nature of the cell. Once multiplied (or amplified), the target DNA can be detected through a variety of methods.

PCR-based methods eliminate the need to culture organisms for detection, and remedy shortcomings of traditional techniques by allowing rapid, sensitive, and specific identification of the pathogens of concern rather than indicator organisms. Traditional culturing can take anywhere from 3 to 7 days, whereas real-time PCR can often be run in 10 to 24 h (Yang and Rothman 2004).

Table 4 Example Case of Airborne Fungi in Building and Outdoor Air

Location	CFU/m ³	Rank Order Taxa
Outdoors	210	<i>Cladosporium</i> > <i>Fusarium</i> > <i>Epicoccum</i> > <i>Aspergillus</i>
Complainant office #1	2500	<i>Tritirachium</i> > <i>Aspergillus</i> > <i>Cladosporium</i>
Complainant office #2	3000	<i>Tritirachium</i> > <i>Aspergillus</i> > <i>Cladosporium</i>

Notes: CFU/m³ = colony-forming units per cubic metre of air. Culture media, for this example, was malt extract agar (ACGIH 1989).

U.S. Environmental Protection Agency (EPA) and Department of Housing and Urban Development (HUD) researchers have developed a metric called the environmental relative moldiness index (ERMI) to objectively describe the home mold burden (Vesper et al. 2007). A DNA-based analysis called mold-specific quantitative PCR (MSQPCR) of 36 molds, including 26 species associated with homes with water damage and 10 found in homes independent of water damage, forms the basis of the ERMI.

Controlling Exposures to Particulate Matter

Control of airborne particulate levels may be achieved by one of four methods:

- Reduction of source emissions
- Capture of emissions at the source using local exhaust
- Dilution using mechanical ventilation
- Removal from ventilation air by filtration

Of these, filtration is of particular importance to HVAC&R. This topic is covered in detail in Chapter 29 in the 2016 *ASHRAE Handbook—HVAC Systems and Equipment*.

Control of bioaerosols is a complex issue because of their capacity for growth and dispersion. However, in general, particulate removal devices and controls are effective in collecting and removing bioaerosols, including allergens (Foarde et al. 1994). Control may also be achieved by ultraviolet irradiation, as described in Chapter 60 of the 2015 *ASHRAE Handbook—HVAC Applications*.

3. GASEOUS CONTAMINANTS

The terms **gas** and **vapor** are both used to describe the gaseous state of a substance. *Gas* is the correct term for describing any pure substance or mixture that naturally exists in the gaseous state at normal atmospheric conditions. That is, its vapor pressure is greater than ambient pressure at ambient temperature. Examples are oxygen, helium, ammonia, and nitrogen. *Vapor* is used to describe a substance in the gaseous state whose natural state is a liquid or solid at normal atmospheric conditions. The vapor pressure is below ambient pressure at ambient temperature. Examples include benzene, carbon tetrachloride, and water. Differences between the two classes reflect their preferred states:

- For a strong source, the concentration of a gas in air in a confined space can rise above one atmosphere. Thus, even nontoxic gases can be lethal if they completely fill a space, displacing the oxygen necessary for survival.
- Vapors can never exceed their saturated vapor pressure in air. The most familiar example of a vapor is water, with relative humidity expressing the air concentration as a percentage of the saturated vapor pressure.
- Vapors, because their natural state is liquid or solid (low vapor pressure), tend to condense on surfaces and be adsorbed.

Gaseous contaminants can also usefully be divided into organic and inorganic types. **Organic** compounds include all chemicals based on a skeleton of carbon atoms. Because carbon atoms easily combine to form chain, branched, and ring structures, there is a wide variety of organic compounds. Despite the variety, they have similarities that can be used in sampling, analysis, and removal. Chemists subclassify organic compounds based on families having similar structure and predictable properties. Organic gaseous contaminants include gases such as methane, but the majority are vapors.

All other gaseous contaminants are classified as **inorganic**. Most inorganic air contaminants of interest to ventilation engineers are gases (mercury is an important exception). Major chemical families of inorganic and organic gaseous contaminants, with examples of specific compounds, are shown in Table 5, along with information

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